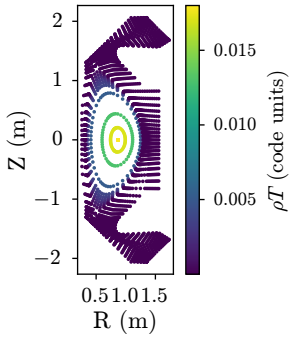
MAST-U OpsSplit with L-SplineFNO

JOREK model 303 — axisymmetric $n = 0$ — pre-pilot study

Proposed partial replacement. Learn density and pressure advection/compression while retaining magnetic and momentum dynamics numerically. Test autonomous generalisation within one fixed MAST-U equilibrium and native finite-element mesh.

(a) Pinned equilibrium nodes



(b) Proposed hybrid update

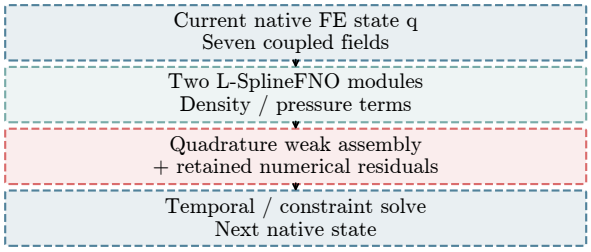


Figure 1: Pinned upstream equilibrium (2876 nodes; colour ρT in code units) and proposed hybrid step. The figure is not a new run; the native numerical bridge remains unvalidated.

1 Axisymmetric reduced-MHD balances

Use JOREK notation $p = \rho T$, $\partial_\phi = 0$ and the case setting $\tau_{IC} = 0$. The field ansatz is

$$\begin{aligned} \mathbf{B} &= \nabla\psi \times \nabla\phi + F_0 \nabla\phi, \\ \mathbf{v} &= -R^2 \nabla u \times \nabla\phi + V_\parallel \mathbf{B}. \end{aligned}$$

Selected balances and constraints are

$$\begin{aligned} \partial_t \rho &= \underbrace{-\nabla\cdot(\rho\mathbf{v})}_{\mathcal{A}_\rho(q)} + \mathcal{R}_\rho(q), \\ \partial_t p &= \underbrace{-\mathbf{v}\cdot\nabla p - \gamma p \nabla\cdot\mathbf{v}}_{\mathcal{A}_p(q)} + \mathcal{R}_p(q), \\ \partial_t \psi &= R[\psi, u] + \mathcal{R}_\psi(q), \\ z_j &= \Delta^*\psi, \qquad w = \Delta_{\mathrm{pol}} u. \end{aligned} \tag{1}$$

Here $q = (\psi, u, z_j, w, \rho, T, V_\parallel)$ and $[a, b] = a_R b_Z - a_Z b_R$. The remainders retain transport, sources, heating and resistivity/current drive. Momentum, Lorentz/pressure forces and native temporal/weak operators stay numerical.

2 Physical terms selected for learning

For $n = 0$ and adiabatic index γ, the selected blocks are

$$\begin{aligned} \mathcal{A}_\rho(q) &= \frac{[R^2\rho, u] - [\rho V_\parallel, \psi]}{R}, \\ \mathcal{A}_p(q) &= R[p, u] + 2\gamma p\, u_Z \\ &\quad - \frac{V_\parallel [p, \psi] + \gamma p [V_\parallel, \psi]}{R}. \end{aligned} \tag{2}$$

3 Proposed neural-operator replacement

Two independent L-SplineFNOs replace $\mathcal{A}_\rho \mapsto \mathcal{N}_{\theta_\rho}$ and $\mathcal{A}_p \mapsto \mathcal{N}_{\theta_p}$:

$$\begin{aligned} \partial_t \rho &= \mathcal{N}_{\theta_\rho}(q) + \mathcal{R}_\rho(q), \\ \partial_t(\rho T) &= \mathcal{N}_{\theta_p}(q) + \mathcal{R}_p(q). \end{aligned} \tag{3}$$

Networks consume current fields at each integrator stage and return scalar RHS terms at native quadrature queries.

With native weak assembly $\mathcal{W}$ and row injections E_ρ, E_p,

$$\begin{aligned} M(q)\dot{q} &= F_{\mathrm{kept}}(q; h) + E_\rho \mathcal{W}[\mathcal{N}_{\theta_\rho}(q)] \\ &\quad + E_p \mathcal{W}[\mathcal{N}_{\theta_p}(q)], \qquad C(q) = 0. \end{aligned} \tag{4}$$

M stays coupled: **a pressure residual is not** $\dot{T}$. Magnetic/momentum dynamics, transport, boundaries, constraints and step-dependent stabilization (h) remain numerical.

4 Evidence and pending comparisons

81 study and 23 spline tests passed. Magnetic-loss diagnostics corrected; full energy and mesh checks remain open. Job **2767720**: **COMPLETED**. Gates A/B/C: pending/pending/pending. **ID/OOD errors, completion and speedup: not measured.**

Equation source. Pinned model303 `mod_elt_matrix_fft.f90`, rows 1–7; local `reports/equations.md`. This is a proposed partial replacement.